

Integrating Azure DevOps and GitHub

IADG | 1 Day



Connect Azure DevOps and GitHub so your team can use the best of both. This one-day course is designed to help you integrate the two platforms across code, work tracking, and pipelines, whether you're standardizing on one, migrating between them, or running both side by side. Through hands-on examples, you'll learn practical ways to link repos, sync work items, and build CI/CD that spans Azure Pipelines and GitHub Actions.

Who should take this course?

This course is for software developers, DevOps engineers, project managers, and IT professionals responsible for managing repositories, automating pipelines, and coordinating work across teams. It's equally valuable for teams already using Azure DevOps or GitHub who want to connect the two and get the most from both. Prior experience with either platform is helpful but not required.

Course content

This course progressively builds your fluency in connecting Azure DevOps and GitHub, from linking repositories and work items to running CI/CD across both platforms. You'll work hands-on through realistic integration scenarios, applying each technique the way a team running both tools would.

1. INTEGRATING FUNDAMENTALS

- Azure DevOps and GitHub: overlap, gaps, and where each is strongest
- Why integrate, and the three patterns: standardize, migrate, or run both
- Choosing your integration strategy
- Authentication models: the Azure Boards app and Workload Identity Federation
- Configuring and verifying the connection
- Troubleshooting common integration issues

2. INTEGRATING WORK TRACKING

- Azure Boards and GitHub: planning in one, building in the other
- Connecting Azure Boards to GitHub repositories
- Linking commits and pull requests with AB#
- Automatic work item state transitions
- Using GitHub Issues alongside Azure Boards
- Delegating work items to GitHub Copilot
- Notifications, status badges, and progress tracking

3. INTEGRATING SOURCE CODE

- Azure Repos and GitHub: choosing where your code lives
- Linking and cross-referencing across platforms
- Migrating repositories with GitHub Enterprise Importer
- Migrating pipelines with GitHub Actions Importer
- Code review and pull requests across both platforms
- Keeping repositories in sync

4. INTEGRATING CI/CD

- Azure Pipelines vs. GitHub Actions: when to use which
- Connecting GitHub repositories to Azure Pipelines
- Building and deploying GitHub code with YAML pipelines
- Pipeline triggers and branch policies
- Calling GitHub Actions from Azure Pipelines (and vice versa)
- Secret-less authentication with Workload Identity Federation (OIDC)
- Multi-stage pipelines and environments

Course Designer

This course was designed by Richard Hundhausen, Microsoft's first Visual Studio ALM/DevOps MVP, a Professional Scrum Trainer, and co-creator of the Nexus framework, with decades helping teams build and ship software. To see other Azure DevOps and GitHub courses, visit www.accentient.com.

